

Project Planning, Work Breakdown Structure and Program Management

WBS	Task Name	QTY
4	TSF2 PH1&2 Dam Wall Raise	
4.1	HDPE Liner ((Based on SERVACO quote)	8775m ²
4.2	Berm Construction (Based on De Simone Rates)	5800m ³
4.3	Trenching for liner installation (Cut and Fill)	720 m ³
4.4	Liner Installation (Based on Citland quote)	8750 m ²
4.5	Manitou rental (estimate based on the Citland offer for PH3)	1 lot
4.6	Labour (estimate based on the Citland offer for PH3)	1 lot
4.7	Relocation of the Evaporator (estimate)	1 lot
4.8	Repairs to the existing liner between PH1 and PH2	1 lot
4.9	Design Cost EPOCH sign off (estimate Hein in negotiation with Epoch)	1 lot
4.10	Contingency cost (allowance)	1 lot

Figure 1. Work Breakdown Structure (WBS) for the TSF2 Phase 1 & 2 Dam Wall Raise. The WBS was developed to define the project scope, organize work packages, estimate quantities and costs, and manage procurement planning, budgeting and construction execution.

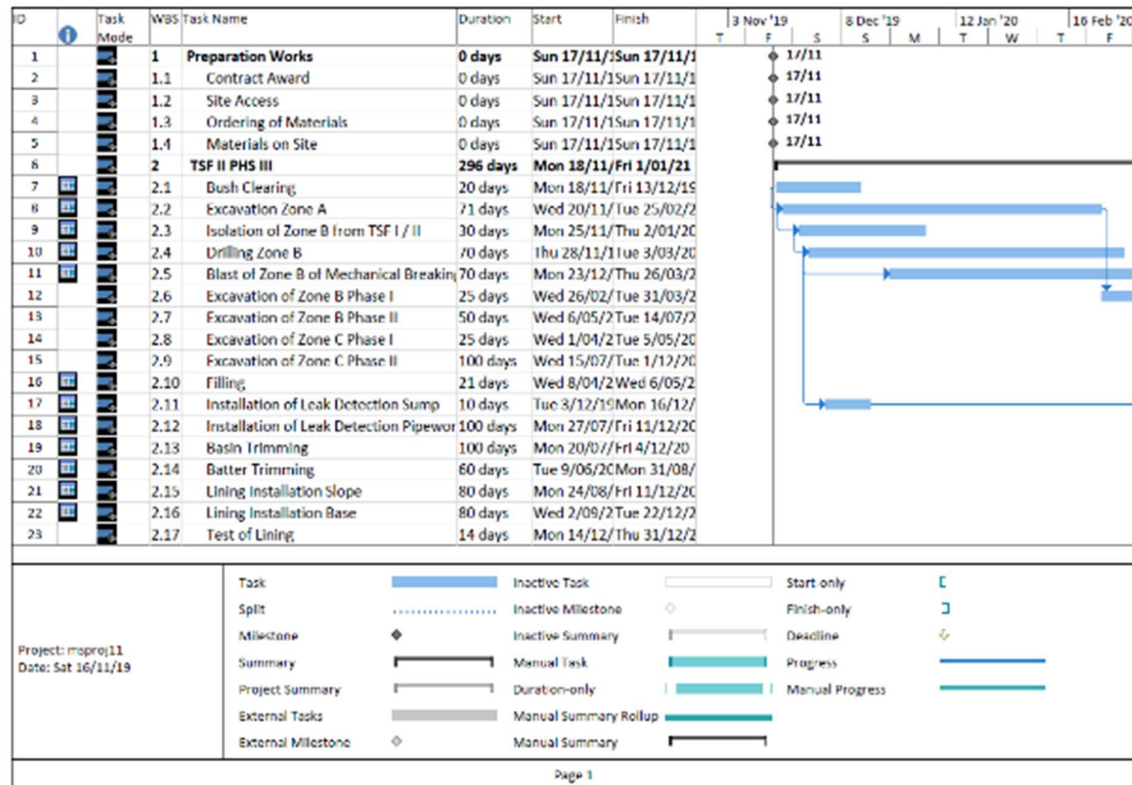


Figure 2. Construction program developed in Microsoft Project for TSF2 Phase III. The program illustrates the planned sequence of construction activities, task dependencies, procurement milestones and commissioning, providing the basis for coordinating engineering, procurement and construction activities throughout project delivery.